

PROCEDURE - function and loops : program using PL/SQL
 procedure / functions & loops

AIM:- To implement PL/SQL procedures, functions and loops on number theory and business scenarios

1. Simple PL/SQL :- program (static input)

DECLARE

message VARCHAR(20) := 'Booking closed';

BEGIN

dbms_output.put_line(message);

END;

OUTPUT:

Booking closed

2. Condition Statement (dynamic input)

DECLARE

hid NUMBER(3) := 100;

BEGIN

IF (hid = 10) THEN

dbms_output.put_line('value of hid is 10');

ELSEIF (hid = 20) THEN

dbms_output.put_line('value of hid is 20');

ELSEIF (hid = 30) THEN

dbms_output.put_line('value of hid is 30');

ELSE

dbms_output.put_line('None of the values matching');

END IF;

dbms_output.put_line('Exact value of hid is ' || hid);

END;

OUTPUT:

None of the value is matching
Exact value of hid is 100

3. Nested Loops Example

DECLARE

hid NUMBER(1);

oid NUMBER(1);

BEGIN

<< OUTER-LOOP >>

for hid IN 1-3 LOOP

<< inner-loops >>

for oid IN 1-3 LOOP

dbms_output.put_line ('hid is: || hid || and oid is: || oid ||');

END LOOP inner-loop;

END LOOP Outer-loop;

END;

OUTPUT:

hid is: 1 and oid is: 1

hid is: 1 and oid is: 2

hid is: 1 and oid is: 3

hid is: 2 and oid is: 1

hid is: 2 and oid is: 2

hid is: 2 and oid is: 3

hid is: 3 and oid is: 1

hid is: 3 and oid is: 2

hid is: 3 and oid is: 3

4. PROCEDURE Example

CREATE OR REPLACE PROCEDURE booking_status
(cid IN NUMBER)

IS
BEGIN


```

If C-id > 200 THEN
    dbms_output.put_line('No booking available');
ELSE
    dbms_output.put_line('Booking open');
END If;

END

```

Execution

```

BEGIN
    booking := status(150);
    -booking - status(250);
END;

```

Output:

Booking open
 No Booking available
 PL SQL Procedure for loops

Example: Using WHILE loop with cursor
 prime check using while loop

CREATE OR REPLACE PROCEDURE print_prime_customers

(CURSOR COST - CUR IS

SELECT CUSTOMERS id FROM CUSTOMERS;

V-id NUMBER;

V-is - prime BOOLEAN;

V-i NUMBER

BEGIN

OPEN COST - CUR;

LOOP FOR COST - CUR INTO V-id;

EXIT THEN COST - (0.8% NOT FOUND)

IF V-ID < 2 THEN

V-IS-PRIME := FALSE;

ELSE

V-IS-PRIME := TRUE;

V-I := 2;

WHILE V-I < TRUNC(SQRT(V-ID)) LOOP

IF MOD(V-ID, V-I) = 0 THEN

V-IS-PRIME := FALSE;

EXIT;

END IF;

V-I := V-I + 1;

END LOOP;

END IF;

IF V-IS-PRIME THEN

DBMS-OUTPUT-PUT-LINE('prime customer
ID: || V-ID');

END IF;

END LOOP;

CLOSE CUST-CUR;

END;

The procedure checks all customer IDs in the table and prints the prime ones using a WHILE loop

EXAMPLE: USING FOR LOOP FOR FIRST N PRIME NUMBERS

CREATE OR REPLACE PROCEDURE print_first_n_primes(n number) IS

V_NUM NUMBER := 2;
V_COUNT NUMBER := 0;
V_IS_PRIME BOOLEAN;

BEGIN

WHILE V_COUNT <= N LOOP
V_IS_PRIME := TRUE;

FOR I IN 2 THRU (SQRT(V_NUM)) LOOP
IF MOD(V_NUM, I) = 0 THEN
V_IS_PRIME := FALSE;

EXIT;
END IF;
END LOOP;

IF V_IS_PRIME THEN

DBMS_OUTPUT.PUT_LINE('Prime: ' || V_NUM);

V_COUNT := V_COUNT + 1;
END IF;

V_NUM := V_NUM + 1;
END LOOP;

END;

This procedure prints the first N prime numbers using a for loop.

```
BEGIN  
  print, first = n - primes(10);  
END;
```

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RESULT: Thus, the procedure function and loops program using PL/SQL procedures, functions & loop are executed successfully.